Store RFID

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Overview

Handheld

Through integration with handheld Radio Frequency Identification (RFID) readers, some Store Inventory and Fulfillment operations can be performed using unit-level RFID tags where item barcodes would traditionally be scanned. When a supported flow is enabled for RFID and the reader is connected, a green RFID indicator appears at the top of the screen to notify the user that the flow is operating in RFID mode. These unit-level RFID tags must follow GS1's Serialized Global Trade Identification Number (SGTIN) standard and have 96-bits of memory.

Configuration determines whether a particular flow (i.e., receiving, packing, etc.) should operate in RFID mode. The Store application supports three pre-defined antenna strengths - short, medium and long. For flows that are enabled, a default antenna strength is configured since for some flows a long read range is desired so the maximum number of tags can be read whereas for other flows a shorter range is desired to prevent unwanted tags from being read. Based on permissions, associates can temporarily change the antenna strength within a flow. The temporary override is reset back to the default once they log out of the application. The antenna strength associated with the short, medium and long levels are defined as 5 dBm, 27 dBm and 30 dBm, respectively. If you wish to adjust the out-of-the-box strengths to different values, contact Manhattan support.

Zebra handheld readers that are certified with Store Apple and Android devices are connected through Bluetooth. Refer to the Certified Technology page for additional information. Custom handheld readers are also supported.

When RFID is enabled, a record exists for each unit in the Unit Inventory table.

Note: For customers using Apple devices with handheld readers, the devices must be running iOS / iPadOS 14 or greater. The Store application asks permission to communicate with an RFID reader over Bluetooth when it is first launched for use

Note: Store RFID supports up to 50,000 units per store across 1000 stores in the network.

These are the statuses which an EPC can have in the Unit Inventory table:

- **InBound**: When an item is shipped to a store as part of a shipment for a customer order or as store stock but has not yet been received.
- **Pending Receipt**: Designates a tag which has been read by an overhead reader and matched to an inbound ASN. Applicable only for stores with overhead readers.
- **Available**: The unit exists in the store and is not reserved for a customer order or missing.
- **Reserved**: The unit exists in the store but has been associated with a customer, pullback or transfer order.
- **Missing**: If an available item unit is found to be missing during a store count, the status of the unit is updated to missing. InBound units are also updated to Missing status if the package they belong to is marked as missing.
- **Departed**: The unit status is updated to departed once the ship event occurs for ship to address, pullback or transfer orders, after pickup has occurred for pickup orders, or after Point of Sale records a sale.
- **Unexpected**: This status is used to denote new tag reads from processes such as Store Count or Overhead that do not already exist in the store. New tags read during an initial store load are updated to "Available" status once the count is submitted. New tags read by other processes such as store count or overhead remain in "Unexpected" status until they are read during a functional process such as picking, packing, point of sale, etc.
- **Removed**: The unit status is updated to "Removed" when a unit is adjusted using a subtraction adjustment type.

Purge

Unit Inventory records in "Missing" status, which are untouched for 30 days, are purged.

Unit Inventory records in "Departed" or "Removed" statuses, which are untouched for 7 days, are purged.

Overhead

In addition to the handheld-based RFID functionality described above, stores with overhead RFID readers can leverage those readers to:

- Store count - Automatically consider units which were recently detected by overhead reader as counted
- Receive inventory - Overhead readers can be used to "pre-receive" inventory for a store associate to review, reducing the reliance handheld readers to get inventory into the store.
- View Inventory on Map - Display all units of an item on a map of the store based on coordinates sent by the overhead reader.
- Pick path optimization - Calculate the best pick path sequence for a picklist based on unit coordinates sent by the overhead reader. The associate can view a map which shows the travel sequence.

For more details on how the Store application can work with overhead RFID readers, please refer to the Overhead RFID - Pick Path Optimization & View Inventory section.

Note: If overhead functionality is used, it must be used in conjunction with handheld flows since not every flow is enabled for overhead readers.

Enablement

If the RFID configuration is enabled for a store function (see Configuration section below), the store operates in RFID mode. If RFID is enabled for a flow (such as packing or receiving), in most occassions store associates must capture unit-level uniqueness by either reading the RFID tag, scanning the Electronic Product Code (EPC) barcode or scanning the SGTIN barcode. Barcodes which only have item-level uniqueness (i.e., item ID, UPC, EAN, GTIN-14) are not allowed in order to retain integrity in the Unit Inventory database.

The GTIN-14 (Global Trade Identification Number) must also be defined in the Item Master for each item. One GTIN-14 value should be defined for each unique UPC (or EAN) barcode that is tied to the item. For example,

- If item ID 123 has only one associated UPC barcode 012345678905 that is defined in the Item Codes entity, then one GTIN-14 value should be defined in the Item Codes entity and would have value 00012345678905 (UPC plus 2 preceding zeros to convert to a 14-digit number.)
- If item ID 456 has two associated EAN barcodes (5701234567899 and 5712234567899), then one GTIN-14 value must be defined for each EAN value in the Item Codes entity (with values 05701234567899 and 05712234567899, respectively, to convert the 13-digit EANs to 14-digit GTIN-14s.)

Tagged and Non-Tagged Items

Store functions which can be enabled for RFID such as (picking, packing, pullbacks, etc.) expect that all items in the store are RFID tagged, unless the function is specifically mentioned in the list below. The functions below reference the "Is RFID Tagged" attribute in the Item Master. If the attribute is set to true and RFID is configured for the function, the function operates in RFID mode for that item. Items which are marked as true must have a GTIN-14 value defined in the Item Code or Manufacturing entity.

If the attribute is set to false or null, these functions treat the item as non-RFID tracked.

Note that if the attribute is set to true, all units of the item must be tagged.

- Receiving - Supports receipt of both RFID tagged and non-tagged items. See below for further information.
- Store count - Supports stores with a mixture of tagged and non-tagged items. Store Counts sends supply updates for all those units which have an RFID tag on and whose item masters have the required GTIN14 configuration in place, irrespective of whether they have the Is RFID Tagged attribute true or false. See below for further information.
- Adjust inventory - Supports both RFID tagged and non-tagged items. See below for further information.

Note: When a unit's information is persisted in the unit inventory table, the characters in the EPC are persisted in the same case as read. If the EPC contains lowercase, it is read as lowercase and persisted as lowercase in the unit inventory table and vice versa. It is also expected that the case should be the same between the physical tag and the record stored in the unit. Any external feeds to unit inventory, if any, should also be in the same case.

Multiple Organization

- Org Column in Unit Inventory table - The Org that to use to populate an EPC's record in unit inventory table is either updated from the org in the user's context or with the parent org that is configured for location ID in the user's context.
- Unit Inventory MUP UI - If a superuser signs in at parent (brand)level, the EPC can be seen by manually changing the org value in the MUP UI dropdown.

Cross Organization

Stores can ship the ship to store orders, Pullback orders, and Store-initiated transfer orders from one organization's location to another organization's location.

Refer to the shipping process guide for additional information.

Refer to the process guide for additional information about the cross-organization feature.

Handheld

RFID Reader Settings

Manhattan Active® Omni allows retailers to pass certain RFID reader settings to the reader during RFID flows. Below is the list of RFID settings that can be passed by the Store application:

- Link Profile Index
- Session
- Tag Population
- Inventory State
- SL Flag
- Dynamic Power

Out of the box base values are provided as seed data which should work well in most RFID environments. These settings can be configured at the device type (i.e., RFD8500, RFD40) or RFID flow ID level (i.e., Packing, Adjust Inventory, Store Count, etc.) The base seed data applies to all device types and flow IDs.

Refer to the configuration section for more details.

Bluetooth RFID Reader Management

Store associates can use the Store app to manage the Bluetooth RFID readers that they use. They can view details of the connected RFID reader, connect to a reader and

disconnect from the currently connected reader. The Bluetooth readers must be previously paired to the device to enable the Store app to connect to them.

View Connected Device Details

On accessing an RFID-enabled function, if a connection to a Bluetooth reader is successfully established, a green RFID indicator is displayed at the top of the UI. When a store associate taps on an active connection (green RFID indicator), the details of the connected reader including the device name, model, current read range and battery level are displayed. Users can adjust the currently configured reader range (antenna strength) by tapping on the "ADJUST RANGE" button. Store associates can view the connected device details even when the RFID reader is in the barcode mode. When the reader is in barcode mode, the range cannot be adjusted.

Connect and Disconnect

Store associates can tap on the "AVAILABLE DEVICES" button to manage the Bluetooth RFID readers. If a reader is already connected, it is displayed at the top of the list. Reader names are displayed as the first seven characters of the device model, followed by the last four characters of the device's serial number. Store associates can disconnect from the currently connected reader or connect to a reader. If the device is connected to a reader and the user wishes to switch to another reader, it is not necessary to disconnect from the first reader. The user can directly connect to the second reader.

On Android devices, all previously paired RFID readers are listed in the "AVAILABLE DEVICES" popup. If the store associate tries to connect to a reader which is currently unavailable (i.e., out of range or powered off), an error message is displayed. On iOS/iPadOS devices, only the readers which are currently connected to the OS and are available to be connected are displayed in the list.

If the device's screen locks and the associate unlocks the device before the Store app times out, when returning to the Store app the device automatically attempts to reconnect as follows:

- For Android, the device will attempt to reconnect to the same reader.
- For Apple, the device will attempt to reconnect, but it may not be the same reader that they were just connected to based on the way that the Apple OS prioritizes the reader list.

If a reader is not actively connected, a red RFID (inactive) indicator is displayed. The store device's Bluetooth needs to be on to enlist the available readers to the user. When a store

associate taps on an inactive connection (red RFID indicator), a message is displayed if the device's Bluetooth is not enabled.

"ADJUST RANGE", "AVAILABLE DEVICES", "CONNECT", and "DISCONNECT" user actions are protected by permission. Refer to the Store Common UI Resources for more information.

Sublocations

RFID-enabled stores can count and track their inventory by sublocation. Sublocation examples are sales floor, back room and off-site storage. Each of the RFID functions can be configured with a default sublocation that is automatically stamped on a unit when an RFID function is executed. The default sublocation for each function is explained in the respective sections below. Refer to the configuration section for details on how to configure a default.

For stores that track inventory by sublocation, the View Inventory UI obtains some data from the Unit Inventory table rather than Inventory Supply. Refer to the View Inventory process guide for additional information.

RFID Reader Trigger Behavior

For all SIF handheld flows except Store Count, tags are read while the reader's RFID trigger is pulled. Once the trigger is released, the reader stops reading and sends the tags it has read to the Store application.

For Store Count, the user pulls and releases the trigger to begin reading tags. Tags are read until the trigger is pulled again which stops the reader. While the reader is reading tags, it sends those tags to the Store application.

Pickup Orders

Store associates can perform picking of pickup orders by reading tags with a handheld RFID reader instead of by scanning item barcodes. Tags read during picking are decoded to determine which item is being picked. However, EPC IDs are not captured during picking but are instead captured as part of a separate capture EPC process. Refer to the Picking Process and Capture RFID Tags guides for additional information, as well as the Configuration section below.

Default Sublocation

If a default sublocation is configured for the Picking - Prepare Pickups flow, it is updated for units processed via the RFID Capture Tags process. Refer to the configuration section for additional information on setting up the default sublocation.

Shipment Orders

The picking and packing of shipment orders can be performed by reading tags with a handheld RFID reader instead of by scanning item barcodes. Tags read during picking are decoded to determine which item is being picked. However, EPC IDs are not captured during picking but are instead captured during packing. Refer to the Picking Process and Packing process guides for additional information, as well as the Configuration section below.

Default Sublocation

If a default sublocation is configured for the Packing flow, it is updated for units processed via RFID packing. Refer to the configuration section for additional information on setting up the default sublocation.

Receiving

Store associates can receive packages for store inventory or customer ship to store orders using a handheld RFID reader. Refer to the Receiving process guide for additional information about RFID-based receiving as well as the Configuration section below.

Default Sublocation

If a default sublocation is configured for the receiving flow, it is updated for units processed via RFID receiving. Note that a single default value exists for receipt of both customer orders and store inventory. Refer to the configuration section for additional information on setting up the default sublocation.

Store Count

Store associates can count inventory in the store using handheld RFID readers and can also leverage overhead readers if the present in the store. Refer to the Store Count guide for additional information about RFID-based store count as well as the Configuration section below.

Counting By Sublocation

With this feature, inventory can be tracked and counted within predefined areas (or sublocations) of the store. If sublocations are defined in the store, the Store count UI displays a list of sublocations for the user to select where they are counting. Counting by sublocation is used in conjunction with Replenishment functionality.

Refer to the configuration section for additional information.

Default Sublocation

The sublocation is updated based on the sublocation that the user is counting. A default sublocation configuration does not exist for store count.

Pullback/Transfer

Store associates can pack pullback/transfer orders using a handheld RFID reader. Refer to the Pullback/Transfer process guide for additional information about RFID-based pullback/transfer order packing as well as the Configuration section below.

Pullback/Transfer - Locate Item

Store associates can utilize a handheld RFID reader if they need assistance locating an item during Pullback/Transfer order packing. Based on permissions, a Locate Item action in Pullback/Transfer order details UI directs the user to the closest unit of an item based on their current location. Visual and audible indicators increase or decrease depending on whether the user is moving closer or farther from the unit. A permission-controlled "next tag" action allows the associate to find another unit of the same item which directs them to the next closest tag based on their current location.

The "Locate Item" feature offers two modes: single tag mode and multi-tag mode.

Once the user initiates the Locate item function, the reader first searches for eligible tags within its range

. This is referred to as multi-tag mode and is accompanied by a rapid beeping sound. Once at least one tag is found, the reader locks onto the tag with the strongest signal and begins showing the percentage from 0 to 100%. This is referred to as single tag mode and is accompanied by a slower-paced beep that changes in frequency as the signal percentage increases (faster) or decreases (slower.).

Note : Tags that exist in the Unit Inventory table in any other status other than "Departed" are eligible for locate item.

Refer to the Configuration section below for more information as well as the Pullbacks & Transfers process guide and for grants refer to Store Inventory Management.

page.

Default Sublocation

If a default sublocation is configured for the pullbacks/transfers flow, it is updated for units processed via RFID pullbacks and transfers. Refer to the configuration section for additional information on setting up the default sublocation.

Store-Initiated Transfer

Store associates can pack store-initiated orders using a handheld RFID reader. Refer to the Store-Initiated Transfer process guide for additional information about RFID-based store-initiated transfer order packing as well as the Configuration section below.

Default Sublocation

If a default sublocation is configured for the store-initiated transfer flow, it is updated for units processed via RFID store-initiated transfers. Refer to the configuration section for additional information on setting up the default sublocation.

Adjust Inventory

Store associates can adjust store inventory using a handheld RFID reader. Refer to the Adjust Inventory process guide for additional information about RFID-based inventory adjustment as well as the Configuration section below.

Default Sublocation

If a default sublocation is configured for the inventory adjustments flow, it is updated for units processed via RFID Adjust Inventory. Refer to the configuration section for additional information on setting up the default sublocation.

Replenishment

With Replenishment functionality, stores that track inventory at the sublocation level can automatically generate activities to replenish the front selling area. Configuration defines the minimum number of units which should be present for an item in the store's selling area. If the inventory drops below this level and the item exists in the non-selling backroom area, a replenishment activity is created.

Store associates use a UI to view the list of items and quantities to replenish to the selling area. Movements are completed using a handheld RFID reader.

Note: The store must count and track inventory at the sublocation level in order to use replenishment functionality.

Refer to the Replenishment guide as well as the Store Count by Sublocation and configuration sections of this document for more information.

Default Sublocation

The sublocation is updated based on the to sublocation of the replenishment tasks. There is no default sublocation configuration.

Picking - Locate Item

Store associates can utilize a handheld RFID reader if they need assistance locating an item during customer order picking. Based on permissions, a Locate Item action in Prepare Pickups and Prepare Shipments directs the user to the closest unit of an item based on their current location. Visual and audible indicators increase or decrease depending on whether the user is moving closer or farther from the unit. A permission-controlled "next tag" action allows the associate to find another unit of the same item which directs them to the next closest tag based on their current location.

The "Locate" feature offers two modes: single tag mode and multi-tag mode.

Once the user initiates the Locate item function, the reader first searches for eligible tags within its range. This is referred to as multi-tag mode and is accompanied by a rapid beeping sound. Once at least one tag is found, the reader locks into the tag with the strongest signal and begins showing the percentage from 0 to 100%. This is referred to as single tag mode and is accompanied by a slower paced beep that changes in frequency as the signal percentage increases (faster) or decreases (slower.)

Note: Tags that exist in the Unit Inventory table in any status other than "Departed" are eligible for Locate Item.

Refer to the Configuration section below for more information as well as the Picking process guide.

View Inventory

Store associates can utilize a handheld RFID reader to select an item's inventory to view instead of scanning a barcode.

Refer to the Configuration section below for more information as well as the View Inventory process guide.

View Inventory - Locate Item

Store associates can utilize a handheld RFID reader if they need assistance locating an item in the store. Based on permissions, a Locate Item action in View Inventory directs the user to the closest unit of an item based on their current location. Visual and audible indicators increase or decrease depending on whether the user is moving closer or farther from the unit. A permission-controlled "next tag" action allows the associate to find another unit of the same item which directs them to the next closest tag based on their current location.

The "Locate" feature offers two modes: single tag mode and multi-tag mode.

When user selects the locate item to find the inventory of an item, reader searches for tags in multi-tag mode with rapid beeping sound. Once it has locked the targeted item unit (EPC), it enters into the single tag mode, accompanied by a slower pace beeping sound. . The search progress indicator then increases from 0% to 100%.

Note: Up to 100 tags can be located for an item. Tags except in the "Departed" status are eligible to be located.

Refer to the Configuration section below for more information as well as the View Inventory process guide.

Refer to the permissions required to enable this feature for a user.

Inventory Runner - Product Requests - Locate Item

Store associates can utilize a handheld RFID reader if they need assistance locating an item in the store. Based on permissions, a Locate Item action in the Product Requests list and details UIs directs the user to the closest unit of an item based on their current location. Visual and audible indicators increase or decrease depending on whether the user is moving closer or farther from the unit. A permission-controlled "next tag" action allows the associate to find another unit of the same item which directs them to the next closest tag based on their current location.

The "Locate" feature offers two modes: single tag mode and multi-tag mode.

When the user selects the locate item to find the inventory of an item, the reader searches for tags in multi-tag mode with a rapid beeping sound. Once it has locked the targeted item unit (EPC), it enters into the single tag mode, accompanied by a slower-paced beeping sound. The search progress indicator then increases from 0% to 100%.

Note: Up to 100 tags can be located for an item. Tags except the "Departed" status are eligible to be located.

Refer to the configuration section below for more information as well as the Product Requests guide.

Refer to the permissions required to enable this feature for a user.

Encode RFID Tags

For situations in which a unit for an item which is RFID tracked arrives at a store untagged or the tag becomes lost or damaged, the “Encode RFID Tags” functionality can be used to encode and apply a new RFID tag to such units.

Default Sublocation

There is no default sublocation updated during the encode RFID process. The user can choose the sublocation from a dropdown in the encode RFID tags screen. Refer to the Encode RFID Tags page for more details on the process, configuration and permissions.

Unit Inventory Outbound Events

Retailers can enable outbound events which are triggered when an update is made to the Unit Inventory table for use by their own systems or custom requirements. A configurable change detection template defines which columns trigger the outbound event.

Refer to the Enable Outbound Events section below for more information.

Exceptions

If a unit-level tag is damaged or missing or the RFID reader is not available, the EPC ID or SGTIN can be scanned by the associate if the barcode is present on the RFID tag.

The unit inventory entity discards the events update if the event's last read date and time is older than the current last read date and time on the unit status. For instance, the unit was shipped at 11:30 AM after being packed at 11:01. While the ship event successfully changed the unit status from "Available" to "Departed" and updated the last read date and time to 11:30 AM, the pack event was stuck in the queue. The unit has already been updated with the most recent event, thus the pack event updates are ignored.

Custom Update - Unit Inventory

Retailers can update the unit inventory records via a custom process. For example, if the retailers are using a custom point of sale and need to update a record in the unit inventory after a counter sale, they can do it by using a custom process.

In the custom process, send a request to the base Store-RFID component to add/update unit inventory.

The Store application does not validate the status of the unit (EPC) when custom updates are sent from an external system, For example, if an update is sent for a tag that is already in the reserved or departed status, there wont be any errors thrown and the update will be successful.

Mandatory fields when sending the custom request.

- EPC ID/SGTIN ID: Either send EPC ID or SGTIN ID in the request.
- Location ID: Store or Location ID where the unit is currently present.
- Item ID: The name of the item that is associated with the EPC/SGTIN.

- Event: The event that triggers to add/update records in the Unit Inventory. Such as pack event, ship event, receive event, etc.
- Event time: Last time when a Unit (RFID tag) was read.

Note: Custom events are not supported.

Refer to ProActive documentation to know more about the custom process.

Life Cycle of Unit Inventory

The table below shows how the status and location of a unit can change over time:

Scenarios	Unit ID	Statu
Store replenishment ASN from DC created for Store 1 and includes the list of EPCs	EPC 1	InBo
Tag read by overhead reader and pre-matched to a pending ASN for store 1 (only applicable for stores with overhead readers)	EPC 1	Pen
Receive in Store 1	EPC 1	Avai
Pack unit in a customer shipment order	EPC 1	Res
Ship the customer shipment order	EPC 1	Dep
Customer returns the unit to the store (assume return is setup to go back to Available)	EPC 1	Avai
Pack unit in a ship to store order to Store 2	EPC 1	Res
Ship the ship to store order	EPC 1	InBo
Package marked as missed during receiving	EPC 1	Miss
Package was found a few days later and received in Store 2	EPC 1	Avai

Scenarios	Unit ID	Statu
Pack unit in a store-initiated transfer to Store 3	EPC 1	Res
Ship the store-initiated transfer	EPC 1	InBo
Receive in Store 3	EPC 1	Avai
Pack for a pullback order (Destination location is Warehouse)	EPC 1	Res
Ship the pullback order	EPC 1	Dep

Unit Inventory UI

Corporate users or admins with necessary permissions can view records in the Unit Inventory UI for investigation or troubleshooting of a unit-level tag. Users can filter by EPC, SGTIN, store, item ID and inventory status. Details including the status, fulfillment ID, package Id, last store event, disposition, last read date/time and coordinates are shown.

This menu is controlled by a permission. Refer to the Store Common UI Resources page for additional information.

Overhead

The Store application can integrate with third-party overhead RFID solution providers. This integration can yield significant benefits to store operations. For more information on how to enable the overhead RFID integration, refer to the Overhead Configuration section below.

Overhead Events

The overhead RFID solution sends information about each unit's latest location in the store. This unit-level EPC information is regularly passed on to Manhattan Active Omni where it is processed and stored.

Below are the key RFID Unit Inventory attributes related to overhead:

- X Coordinate - X coordinate value of location of the unit
- Y Coordinate - Y coordinate value of location of the unit

- Zone ID - Reference Zone ID of the location where the unit currently resides. Optional.
- Confidence - Confidence in the overhead RFID read. The higher the confidence, the higher the accuracy of the location coordinates captured by the read. Defaulted to 100 if not supplied in the event. Optional.

Note: Movement updates for a unit should be sent from the overhead system only if the unit has moved 1 meter or more, and inactivity updates for a unit should be sent no more frequently than every 4 hours. Manhattan Active Omni supports receiving overhead updates for up to 10 million units per hour across all stores in the network.

Note: Zone ID is optional and is used only for clustering. For plotting the View Inventory Map and the Pick Path Sequence, Manhattan Active® Omni Store Order Fulfillment leverages only the X and Y coordinates received as part of the overhead RFID events. If Zone ID is supplied as part of the overhead RFID events, the retailer's overhead RFID solution provider should ensure its accuracy for clustering to work as expected.

Overhead Receiving

As overhead events are received, MAO checks if the tag matches to an expected package receipt for the store (regardless of whether a tag feed is in place.) Matched tags are updated to "PendingReceipt" status and the corresponding package(s) are pre-loaded on the Receiving UI for store associates to review and confirm receipt. Handheld RFID readers can be used to capture any tags that were not detected by the overhead reader.

Similar to standard handheld receiving, associates have the option to discard any pre-loaded packages in the Receive Inventory UI. Any discarded overhead packages will reappear the next time an associate enters the Receive Inventory UI, essentially deferring the receipt until a later time.

Note: Once tags for the package are detected by overhead, it can take up to 5-10 minutes for the package to appear on the Receiving UI. In the meantime, associates always have the ability to load the package on the UI using a handheld reader.

Updates

When an overhead event is received, the tag is updated based on this logic:

- If the tag does not exist in Unit Inventory and matches to an inbound receipt, the record is created and updated to “PendingReceipt” status
- If the tag exists in Unit Inventory in “InBound” status and matches to an inbound receipt, the record is updated to “PendingReceipt” status.

- If the tag does not exist in Unit Inventory and does not match to an inbound receipt, the record is created and updated to “Unexpected” status.

Refer to the Receiving guide for more information on how the receiving process is completed for tags that are read by overhead readers.

Store Count

Refer to the Store Count guide for information on how Store Count can leverage RFID overhead readers.

Pick Path Optimization

With overhead data, picking efficiency can be significantly improved since the picklist is generated based on the latest known location of ordered items. The picklist is created with an optimized pick sequence and displays the travel path on a map to the associate. The map contains all the pick locations to be traversed in relation to the starting point, which is configurable.

Refer to the Picking process guide for more details on how Picking functionality works in the Store application.

View on Map

The Picking, View Inventory, and Product Requests UIs allow an associate to view the latest known location of item inventory on a store layout map. This can be used to find the nearest unit of an item, or to locate a unit which isn't where it should be.

For more details on the View Inventory functionality, refer to its process guide.

For more details on the Product Requests functionality, refer to its guide.

Clustering

Clustering is an optional functionality which groups nearby points together on the map so that they are represented as a single point. This can be helpful in improving readability of the map, especially on handheld devices. When clustering is enabled, it applies to both Picking (pick path optimization) and View Inventory.

See the clustering example below. Assume the configuration is set to 50 and zone ID is passed in the overhead event.

Before Clustering	After Clustering

• **Point 1 (100, 100)** in Zone 1 - 1 unit of Item 1
• **Point 2 (105, 101)** in Zone 1 - 1 unit of Item 1
• **Point 3 (120, 125)** in Zone 1 - 1 unit of Item 2
• **Point 4 (135, 145)** in Zone 2 - 1 unit of Item 3
	• **Point 1 (100,100)**
	  • 2 units of Item 1 in Zone 1
	  • 1 unit of Item 2 in Zone 1
	• **Point 2 (135,145)** in Zone 2 - 1 unit of Item3

As a result of clustering, the number of distinct points on the map are reduced from 4 to 2, which makes the map less cluttered for the store associates. Even though 1 unit of Item 3 also falls within the clustering range, it is not aggregated as it lies in a different zone.

Note: Clustering logic is based on the clustering range that is defined and the unit's zone. If a zone is not passed in the overhead events, clustering can still occur but will cluster solely based on proximity and will not honor zone boundaries. This may lead to items belonging to different zones being clustered together. **In the example above, if a zone is not supplied as part of the overhead events, all four points will be clustered as one single point.**

Assumptions

The following limitations apply to overhead RFID:

- In the Picking UI, pause and resume is only supported on the same device. A paused pick task must not be resumed on another device.
- If a picklist is printed, items are not grouped by pick locations and appear under the "Others" grouping.
- The non-RFID Pick Path Optimization capability, detailed here, is not supported along with overhead RFID.
- Pick to Slot is not supported with overhead RFID.
- Once a map is generated, it isn't refreshed. Consequently, it's possible that a unit displayed on the map may not be found if it has been picked by another user or moved by a customer.

Phased RFID Enablement

Store count using RFID can be enabled independently from enabling RFID in other store inventory, store fulfillment and point of sale functions. This allows retailers to quickly benefit from the improved accuracy of store counts while rolling out additional RFID-based store functions in parallel in a phased manner as they are ready. Refer to the Store Count guide for additional information.

In this mode, the expected quantities in store count are obtained from Inventory Supply rather than the Unit Inventory table. In the Unit Inventory table, each unit's status is only updated during store count and has either status "Available" or "Missing". The capture of RFID tags during store operations such as picking, packing, receiving and POS transactions is optional and only occurs once the store has enabled RFID for each of those functions.

Refer to the configuration section for additional information.

Order Lookup to support RFID tag-based Returns

If a retailer intends to support RFID tag based returns, Store RFID has the capability to make a note of the Order IDs when an RFID tagged unit is fulfilled by / sold at a store. When this support is enabled, whenever a "Departed" event is generated for any Pickup at Store, Ship to Store, Ship to Address, Same Day Delivery, or Store Sale orders, the departed unit's RFID tag details are saved along with its Order ID.

Order lookup for orders fulfilled from DC is also supported. Orders that are shipped from the DC have an EPC ID shared on the ship event. This information is then saved into the fulfillment detail information on the order. Order can be looked up using EPC ID when the order is shipped from the DC in the Order component by using the /order/api/order/fulfillmentevent/receive API. This API allows you to receive fulfillment events, such as the shipment of an order, and update the status of the order and its fulfillment details accordingly. When you receive a shipment event, you can use the EPC ID included in the event to look up (search) the order and its associated fulfillment details.

Note: Store RFID order lookup capability records only those units which are fulfilled by / sold at an RFID-enabled store or are shipped from DCs. Orders fulfilled by third party vendors are not recorded.

Tip: Currently, the Manhattan Active suite of Omni channel commerce solutions does not support RFID tag based returns out of the box. Order Lookup capability is, however, available for integration with external applications.

Additionally, an API is available for integration with external applications, which accepts upto ten returned units' RFID tags' EPC or SGTIN IDs and returns their respective Order IDs.

The calling application must then integrate with the Order component to confirm if the Order IDs returned by the API are eligible for return and proceed with the return order creation process accordingly.

Tip: As with any other returns, Store Associates are advised to verify that the customer's details match with those mentioned in the Order fetched using the returned items' RFID tag details.

All records older than 365 days are purged and RFID tag-based returns cannot be processed for such orders. If the retailer's policy permits returns beyond this period, such returns should be processed using the customer or order details.

Refer to the configuration section for additional information.

Refer to configuration of Order Lookup for orders fulfilled from DC: Store RFID (helpdocsonline.com)

Decode Custom EPC Tags

The Store application allows retailers to decode tags whose encoding does not follow GS1 standards. By defining a regEx pattern and an itemCode offset for decoding the custom EPC/SGTIN, the Store RFID functions can decode the custom encoded tags.

If the custom decode fails during the decoding process, the system attempts to decode the tag using the base decoding flow (based on GS1 standards.)

Refer to the Configuration section above to understand more about how the regEx and offset values can be set.

Note: Encoding using custom encoded is not supported. Only custom tag decoding is supported.

Note: The custom decode logic supports parsing tags that have a decimal format. Parsing of tags encoded in hexadecimal is not supported.

Configuration

Ensure that the purposes listed below have been copied from the base profile to your custom profile.

Profile purposes to copy from base to custom profile	sto::storeConfig, srf::rfConfig, dvc::rfidReaders

The RFID configuration below can be enabled to all stores in the organization, or can be overridden at the store profile type or location level. Refer to this Config UI guide for additional details, and the profile purposes which are listed at the bottom of this page.

Handheld Configuration

RFID for Packing

If this flag is enabled, store associates pack customer shipment orders using handheld RFID readers. The default RFID antenna range for packing is set as "Short", but can be changed if desired. Based on permissions, store associates can also change the default range on the fly.

The default sublocation is set as null and can be changed to one of the configured sublocations.

This configuration is available in the Store RFID Parameters UI and is disabled by default.

Configuration Name: RFID enabled flows. Flow Name: Packing. The default value is set as false.

The default Read Range is "Short". Change the read range if needed.

The default Sublocation is null. It must be configured with one of the configured sublocation values, if subocation is configured. Else leave it blank.

RFID for Receiving

If this flag is enabled, store associates can receive using handheld RFID readers. The default RFID antenna range for receiving is set as "Medium", but can be changed if desired. Based on permissions, store associates can also change the default range on the fly.

The default sublocation is set as null and can be changed to one of the configured sublocations.

This configuration is available in the Store RFID Parameters UI and is disabled by default.

Configuration Name: RFID enabled flows. Flow Name: Receiving. The default value is set as false.

The default Read Range is "Medium". Change the read range if needed.

The default Sublocation is null. It must be configured with one of the configured sublocation values, if sublocation is configured. Else leave it blank.

RFID for Pickup Orders

If this flag is enabled, store associates can fulfill BOPIS and Same Day Delivery orders using handheld RFID readers. The default RFID antenna range for capturing RFID tags of units picked as part of BOPIS and Same Day Delivery orders is set as "Short", but can be changed if desired. Based on permissions, store associates can also change the default range on the fly.

The default sublocation is set as null and can be changed to one of the configured sublocations.

This configuration is available in the Store RFID Parameters UI and is disabled by default.

Configuration Name: RFID enabled flows. Flow Name: Picking - Prepare Pickups. The default value is set as false.

The default Read Range is "Short". Change the read range if needed.

The default Sublocation is null. It must be configured with one of the configured sublocation values, if sublocation is configured. Else leave it blank.

RFID for Shipment Orders

If this flag is enabled, store associates can fulfill shipment orders using handheld RFID readers. The default RFID antenna range for capturing RFID tags of units picked as part shipment orders is set as "Short", but can be changed if desired. Based on permissions, store associates can also change the default range on the fly.

The default sublocation is set as null and can be changed to one of the configured sublocations.

This configuration is available in the Store RFID Parameters UI and is disabled by default.

Configuration Name: RFID enabled flows. Flow Name: Picking - Prepare Shipments. The default value is set as false.

The default Read Range is "Short". Change the read range if needed.

The default Sublocation is null. It must be configured with one of the configured sublocation values, if sublocation is configured. Else leave it blank.

RFID for Store Count

If this flag is enabled, store associates can count the store's inventory using handheld RFID readers. The default RFID antenna range for store count is set as "Long", but can be changed if desired. Based on permissions, store associates can also change the default range on the fly.

The default sublocation is not used for this flow and can be kept as null.

This configuration is available in the Store RFID Parameters UI and is disabled by default.

Configuration Name: RFID enabled flows. Flow Name: Store Count. The default value is set as false.

The default Read Range is "Long". Change the read range if needed.

RFID for Pullback/Transfer

If this flag is enabled, store associates can pack pullback/transfer orders using handheld RFID readers. The default RFID antenna range for pullback/transfer order packing is set as "Short", but can be changed if desired. Based on permissions, store associates can also change the default range on the fly.

The default sublocation is set as null and can be changed to one of the configured sublocations.

This configuration is available in the Store RFID Parameters UI and is disabled by default.

Configuration Name: RFID enabled flows. Flow Name: Pullback/Transfer. The default value is set as false.

The default Read Range is "Short". Change the read range if needed.

The default Sublocation is null. It must be configured with one of the configured sublocation values, if sublocation is configured. Else leave it blank.

RFID for Store-Initiated Transfers

If this flag is enabled, store associates can pack store-initiated transfer orders using handheld RFID readers. The default RFID antenna range for store-initiated transfer order packing is set as "Short", but can be changed if desired. Based on permissions, store associates can also change the default range on the fly.

The default sublocation is set as null and can be changed to one of the configured sublocations.

This configuration is available in the Store RFID Parameters UI and is disabled by default.

Configuration Name: RFID enabled flows. Flow Name: Store Initiated Transfer. The default value is set as false.

The default Read Range is "Short". Change the read range if needed.

The default sublocation is null. It must be configured with one of the configured sublocation values, if sublocation is configured. Else leave it blank.

RFID for Adjust Inventory

If this flag is enabled, store associates can adjust store inventory using handheld RFID readers. The default RFID antenna range for inventory adjustment is set as "Short", but can be changed if desired. Store associates cannot change the range on the fly.

The default sublocation is set as null and can be changed to one of the configured sublocations.

This configuration is available in the Store RFID Parameters UI and is disabled by default.

Configuration Name: RFID enabled flows. Flow Name: Adjust Inventory. The default value is set as false.

The default Read Range is "Short". Change the read range if needed.

The default sublocation is null. It must be configured with one of the configured sublocation values, if sublocation is configured. Else leave it blank.

RFID for View Inventory

If this flag is enabled, store associates can view the item's inventory using handheld RFID readers. The default RFID antenna range for inventory adjustment is set as "Short", but can be changed if desired. Based on permissions, store associates can also change the default range on the fly.

The default sublocation is not used for this flow and can be kept as null.

This configuration is available in the Store RFID Parameters UI and is disabled by default.

Configuration Name: RFID enabled flows. Flow Name: View Inventory. The default value is set as false.

The default Read Range is "Short". Change the read range if needed.

The default sublocation is null. It must be configured with one of the configured sublocation values, if sublocation is configured. Else leave it blank.

Enable decoding of the item's unit barcode

If this flag is enabled, store associates can scan the Electronic Product Code (EPC ID) or Serialized Global Trade Identification Number (SGTIN) in all SIF UIs which accept item ID (regardless of whether RFID is enabled for the flow.) The Store application decodes the

inputed EPC or SGTIN value to determine the associated item ID. Enabling this flag can be useful if the EPC ID or SGTIN barcode is printed on the RFID tag in lieu of a UPC barcode, or if the retailer wishes that associates always scan one barcode throughout the application.

This configuration is available in the Store RFID Parameters UI and is disabled by default.

Configuration Name: Enable decoding of unit barcode.

RFID Locate Item for Prepare Pickups

If this flag is enabled, store associates can use the handheld RFID reader to locate items during picking of Pickup orders. Note that for the locate function, the range is automatically set to the maximum allowed value by Zebra. So the range value for locate functions is not used and can be ignored.

This configuration is available in the Store RFID Parameters UI and is disabled by default.

Configuration Name: RFID Locate Item. Flow Name: Picking - Prepare Pickups. The default value is set as false.

RFID Locate Item for Prepare Shipments

If this flag is enabled, store associates can use the handheld RFID reader to locate items during picking of Shipment orders. Note that for the locate function, the range is automatically set to the maximum allowed value by Zebra. So the range value for locate functions is not used and can be ignored.

This configuration is available in the Store RFID Parameters UI and is disabled by default.

Configuration Name: RFID Locate Item. Flow Name: Picking - Prepare Shipments. The default value is set as false.

RFID Locate Item for View Inventory

If this flag is enabled, store associates can use the handheld RFID reader to locate items requested as part of Inventory Runner - Product Requests. Note that for the locate function, the range is automatically set to the maximum allowed value by Zebra. So the range value for locate functions is not used and can be ignored.

This configuration is available in the Store RFID Parameters UI and is disabled by default.

Configuration Name: RFID Locate Item. Flow Name: View Inventory. The default value is set as false.

RFID Locate Item for Product Requests

If this flag is enabled, store associates can use the handheld RFID reader to locate items . Note that for the locate function, the range is automatically set to the maximum allowed value by Zebra. So the range value for locate functions is not used and can be ignored.

This configuration is available in the Store RFID Parameters UI and is disabled by default.

Configuration Name: RFID Locate Item. Flow Name: Product Requests. The default value is set as false.

RFID Locate Item for Pullbacks & Transfers

If this flag is enabled, store associates can use the handheld RFID reader to locate items during packing of Pullbacks & Transfers. Note that for the locate function, the range is automatically set to the maximum allowed value by Zebra. So the range value for locate functions is not used and can be ignored.

This configuration is available in the Store RFID Parameters UI and is disabled by default.

Configuration Name: RFID Locate Item. Flow Name: Pullbacks & Transfers . The default value is set as false.

Phased RFID Enablement

This allows a retailer to begin perform RFID store counts before enabling RFID across their other store functions. If this flag is enabled, the expected count for store count is obtained from inventory supply rather than the Unit Inventory table. This flag is intended only for customers who wish to begin using Store Count but do not have all of their store processes enabled for RFID. Once all processes are enabled for RFID, this flag should be disabled so that the typical store count from the Unit Inventory table occurs.

This configuration is available in the Store RFID Configuration UI and is disabled by default.

Configuration Name: Load Expected Value From Supply. The default value is set as false.

Replenishment

If this flag is enabled, store associates can use the handheld RFID reader to replenish inventory from a non-selling area to a selling area. The default RFID antenna range for RFID Replenishment is set as "Short", but it can be changed if desired. Based on permissions, store associates can also change the default range on the fly.

The default sublocation is not used for this flow and can be kept as null.

This configuration is available in the Store RFID Parameters UI and is disabled by default.

Counting by Sublocation

If sublocations are defined for the store, Store Count operates in sublocation mode. Sublocations can be created at organization, store or store profile level. When a sublocation is created, a sublocation type ('Selling Area' and 'Non-Selling Area') must be assigned.

This configuration is available in the sublocation UI and is disabled by default.

Overhead Configuration

Use overhead RFID data to plot Store Inventory on map

This configuration controls whether the overhead RFID integration is enabled. This flag is available in the **Details tab of the Store Fulfillment Parameters** and must be enabled to leverage the pick path optimization or view on map functionalities.

Configuration Name: Use overhead RFID data to plot the Store Inventory on a map. The default value is set as false.

Store Layout

To enable the pick path optimization and the view inventory map overhead RFID functions in the Manhattan Active® Omni Store app, every store where these functions are to be enabled should have its layout configured. As a pre-requisite, the store layout configured in MA Omni should be in sync with the overhead reader so that the coordinates relayed by the overhead are in alignment with the layout. There are two ways of defining a store layout in MA Omni.

Option 1 - Coordinate-based store layout

A typical store layout consists of zones (sections within a store) where the inventory is present. The shape of a store and its zones are defined by a collection of points indicated by their coordinates. This section explains how a store layout can be configured by defining its coordinates.

The following are the key attributes defined as part of a store layout:

- LocationId - Location ID of the store for which the layout is being defined
- StoreLayoutId - Unique identifier of the layout
- Zone.ParentZoneId - Defines the nesting of zones which is used to place a zone inside another one. A zone without a parent represents the outer layout of a store. Only one zone without a parent is allowed per store.

- Zone.Point.XCoordinate - The value of x coordinate which represents a point in the zone
- Zone.Point.YCoordinate - The value of y coordinate which represents a point in the zone
- Zone.Point.SequenceNumber - Define the sequence of the points of the zone in a clockwise or counter-clockwise manner, and use the same consistent manner for all zones in the layout. As an example, let's consider a Coats and Jackets zone which extends from A (0, 0), B (200, 0), C (200, 200), and D (0, 200). If you choose to sequence the points in a counter-clockwise direction, the sequence of the points A, B, C, D should be 1, 2, 3, 4. If choosing the clockwise direction, the sequence numbers of the points A, B, C, D should be 1, 4, 3, 2.
- ZoneId - Name of the zone which is displayed to the associates on the store layout map (i.e., Dressing Room 1, Men's Apparel, etc.)

Tip: Defining co-ordinate points in multiples of 100 for each unit, helps with better rendering of the map. For example, a store can be setup as A(0,0), B(0,1000), C (1000,1000), D (1000,0). Men's zone can be A (200,0), B (200,200), C (400, 200), D (400,0). A shirt can be located at (250, 100)

There is no UI to enter Store Layout configuration. Use the POST API below to configure store layouts.

{{url}}/storerfid/api/storerfid/rfStoreLayout/save

Refer to the "Set up store layout" page of the Store RFID integration guide in the Manhattan Developer Hub for a sample payload which can be used to create a new Store Layout in the Store application with the help of its coordinates.

Option 2 - Upload a store layout image

This section explains how a store layout file can be directly and seamlessly imported into Manhattan Active® Omni. For this functionality to work, the file should adhere to the following guidelines:

1. It should be an SVG file with a recommended size of 500 KB and a maximum size of 1 MB.
2. It should have a view box, comprising both maxima and minima, defined for both the X and Y axes.
3. It should be zipped

4. It should be configured individually for each store. As a corollary, a zipped folder should comprise only one SVG file.

Note: If a Mac machine is used to create the zip file, it must be ensured that the MACOSX file is removed from the zipped folder before uploading it into Manhattan Active® Omni.

The following information is needed to upload a store layout.

- LocationId - The unique identifier for a store that is configured in MA Omni
- File - A zipped folder comprising the store layout file in the SVG format upto a maximum of 1 MB in size

There is no UI to upload the store layout. Use the POST API below to upload the required store layouts.

{{url}}/storerfid/api/storerfid/storeLayout/upload

Refer to the "Upload store layout image" page of the Store RFID integration guide in the Manhattan Developer Hub for detailed information on how a store layout can be uploaded. Additionally, refer to the Developer Hub to get to know about the APIs which can be used to validate if an SVG store layout has been uploaded for a given location, to delete an uploaded store layout file's association with its location, and to view the file uploaded for a given location.

Note: Currently, only single floor stores are supported in both the above approaches. Stores with multiple floors are not supported.

Overhead Read Expiry Threshold

Refer to the Store Count guide for configuration information.

Range to be used for clustering in the overhead RFID store layout

This configuration resides in the **Store Overhead RFID Configuration subsection of the Store RFID Configuration**. A value > 0 directs the system to cluster all the locations that fall within the specified range. To disable clustering, set the range to 0 or null.

Configuration Name: Range to be used for clustering in the overhead RFID store layout. The default value is set as null, which implies that clustering is disabled.

Origin of Store Layout

The coordinate relationship that is used by the third-party overhead RFID system must be mirrored in Manhattan Active® Store in order for the inventory data and store layout to display properly on the UI. This configuration in the **header of the Store RFID**

Configuration defines the origin of the map used in the Store application. Supported origins are Top Left and Bottom Left (default for coordinate-based store layouts). Based on the value configured, the store UI manipulates the coordinates in such a way that the orientation of the layout on the vendor's store layout creation UI and the MA Store UI are the same.

Configuration Name: Origin of Store Layout. The default value is Bottom Left for coordinate-based store layouts.

Note: This configuration applies to coordinate-based store layouts only. For SVG image based store layouts, the only origin supported is Top Left. It is imperative to ensure that both the retailer's overhead RFID solution provider and the MA Store app use the origin as the top left corner of the uploaded store layout.

X and Y coordinates of the starting point in the overhead RFID store layout

A configurable location (such as a packing station or POS station) can be used as the starting point of the picker by specifying its X and Y coordinates. This is the point from where the picking is assumed to begin and is one of the critical factors considered in the generation of the optimized pick path which is suggested to the store associate. If these coordinate values are not specified, the values (0,0) are used.

This configuration resides in the **Store Overhead RFID Configuration sub-section of the Store RFID Configuration**.

Configuration Name: X coordinate of the starting point in the overhead RFID store layout. The default value is null, which is internally considered as 0.

Configuration Name: Y coordinate of the starting point in the overhead RFID store layout. The default value is null, which is internally considered as 0.

Order Lookup to support RFID tag-based Returns Config

The following configuration, if enabled, ensures that the units fulfilled by / sold at an RFID enabled store are noted down so that an external application can determine the Order ID(s) associated with the RFID tag(s) fastened on the item(s) being returned. If this configuration is disabled, the departed units aren't recorded and if the provided API is called, a 'feature disabled' validation error is returned. In essence, enabling this configuration is a pre-requisite for facilitating RFID tag-based returns in stores.

This configuration is available in the **Store RFID Configuration** UI and is disabled by default.

Configuration Name: Support Order lookup using RFID tag data

Note: Owing to the large amount of data which may end up getting recorded in the Store RFID component, Manhattan Associates advises that this configuration is enabled only if an external application intends to support RFID tag based returns.

Order Lookup to support orders shipped from DC with an EPC

The below configuration, is applicable for order lookup for the orders fulfilled from DC. If enabled, it ensures that the units fulfilled from a DC are noted down so that an external application can determine the Order ID(s) associated with the RFID tag(s) fastened on the item(s) being shipped. If this configuration is disabled, the shipped units aren't recorded and if the provided API is called, a 'feature disabled' validation error is returned.

Orders that are shipped from the DC have an EPC ID shared on the ship event. This information is then saved into the fulfillment detail information on the order. Order can be looked up using EPC ID when the order is shipped from the DC in the Order component by using the /order/api/order/fulfillmentevent/receive API. This API allows you to receive fulfillment events, such as the shipment of an order, and update the status of the order and its fulfillment details accordingly. When you receive a shipment event, you can use the EPC ID included in the event to look up (search) the order and its associated fulfillment details.

This configuration is available in the **Store RFID Configuration** UI and is disabled by default.

```
Configuration Name: Support Order lookup using RFID tag data
```

RFID Reader Setting Configuration

This configuration is available in the **RFID Reader Settings** UI and is enabled by default with base seed data.

1. If the RFID reader performs well across the various RFID flows using the base seeded RFID reader settings, no configuration is needed.
2. If the RFID reader does not perform as expected using the base seed data, change any of the base settings by editing the record and updating the key value in the UI.
3. To create override configuration records so that a particular device type or flow ID uses different settings than the wildcard records, create new override records with the desired device type or flow ID override values.
    - For device type and flow ID, null represents a wildcard value. Device type is free text and must be entered as "RFD8500" or "RFD40" for the respective Zebra readers. Key values are also free text. Refer to Zebra's documentation for these supported values for Android and iOS, respectively.

Note: All key values are case-insensitive.

Key	Key Value
Link Profile Index	0
Session	1
Tag Population	30
Inventory State	a
slFlag	all
Dynamic Power	enabled

Enable Outbound events

When the status or any fields on the Unit Inventory entity change, an outbound message can be triggered. The outbound event generation is based on the following main configuration:

{{url}}/storerfid/api/eventProducer/eventStrategy/save

{ "ProfileId": "base", "data": [{ "EntityStrategyId": "UnitInventoryStatusStrategy", "IsRuleRequired": false, "EntityId": "UnitInventory", "Active": false, "IsOldValuesRequired": true, "DetectionTemplateId": "UnitInventoryOutboundEventDetectionTemplate", "Component": "com-manh-cp-storerfid", "TemplateId": "UnitInventoryOutboundEventTemplate", "MessageType": "UnitInventoryOutboundEventMSGType" }] }

Attributes of the eventStrategy.json are explained below.

IsRuleRequired	Default value is false If the requirement is to send an event only when the status has a spec When true, this needs to be followed by multiple other configurations executed by the rule framework, etc.,

Active	By default this is disabled. This value should be set to "True" to get the unit inventory entity.
IsOldValuesRequired	This is currently enabled. If a change set with old values is not require
"EntityId": "UnitInventory"	The entity that is generating the events is specified here.
DetectionTemplateId	A template specified here with a set of fields will be used to determin events.

The UnitInventoryOutboundEventDetectionTemplate template determines when to trigger an event. The out of the box base template is listed below and is configured to generate events only when the EPC's status changes. To generate an event based on updates to other columns, add those columns to the detection template or create a separate event strategy for that column as needed.

responseTemplate.json - {{url}}/storerfid/api/fwcore/responseTemplate/save

{ "TemplateId": "UnitInventoryOutboundEventDetectionTemplate", "BaseEntity": "UnitInventory", "TemplateJson": { "UnitId": null, "Status": null } }

The UnitInventoryOutboundEventTemplate template is applied to the message sent to the queue. Only the fields specified in this template will be posted to the queue. The base response template UnitInventoryOutboundEventTemplate that is supplied out of the box is as follows

responseTemplate.json - {{url}}/storerfid/api/fwcore/responseTemplate/save

{ "TemplateId": "UnitInventoryOutboundEventTemplate", "BaseEntity": "UnitInventory", "TemplateJson": { "UnitId": null, "EpcId": null, "ItemId": null, "Sgtin": null, "Status": null, "LocationId": null, "LastReadDateTime": null } }

This specifies the queue to which the message will be published. The base messageType UnitInventoryEventOutboundMSGType is defined as follows

outboundMessageType.json - {{url}}/storerfid/api/ServiceDefinition/outBoundMessageType/save

{ "MessageType": "UnitInventoryOutboundEventMSGType", "Description": "Unit Inventory Outbound Event MSG Type", "Transactional": true, "PersistMessageToMsgStore": false, "OutBoundQueues": [{ "QueueName": "UnitInventoryOutboundEventMSGType" }] }

Tag Decode Configuration

The tag decode configuration is set false by default. If configured, then system expects records for EPC and SGTIN with unique decode IDs. The tag decode configuration consists of the following parameters:

- Decode Id - Unique identifier of the tag. Examples : D001.
- Tag Decode Configuration - Type of the tag, could be either EPC or SGTIN
- RegEx Pattern - Custom tag decode regex pattern to validate the number. Examples - ^[0-9A-Fa-f]+$
- Item Code Offset - Item Barcode offset values i.e. Start and End offsets of RegexPattern of DecodeType. Examples: 2,16. The offset is defined starting 0. For example, for this

Troubleshooting

Issue	Troubleshooting Tip
Overhead RFID functionalities do not work as expected even after enabling the required configuration in the Store Fulfillment Parameters and Store RFID Configuration.	Ensure that the latest app is installed on the device on which the issue is being encountered.
Incorrect EPC is created in unit inventory	Records in unit inventory can be deleted using this API: Endpoint: storerfid/api/storerfid/unitInventory/deleteUnitInventory Copy to clipboardCopy to clipboardCopy to clipboardCopy to clipboardCopy to clipboard **Payload**:

Issue	Troubleshooting Tip
	{ "EpcIds": ["303401BF841E4282540BE42E"] }Copy to clipboardCopy to clipboardCopy to clipboardCopy to clipboardCopy to clipboard For more details, refer to Developer Portal **Resource:** api::storerfid::unitinventory::deleteunitinventory
"Unable to fetch device details" error in the Connected Device panel while trying to view the connected RFID reader details	Please retry. Contact Manhattan Associates if the issue persists.
"Couldn't fetch available devices. Please retry!" error in the Available Devices panel while trying to manage the available RFID readers	Please retry. Contact Manhattan Associates if the issue persists. While the issue is being resolved, you may use the native OS capabilities to connect to or disconnect from a device.
"Couldn't connect to the device. Ensure the device is powered on with	Ensure that the device that you are trying to connect to is switched on and Bluetooth turned on.

Issue	Troubleshooting Tip
Bluetooth enabled." error while trying to connect to a reader in the Available Devices panel	
"Couldn't disconnect. You may retry here or using your OS native capability." while trying to disconnect from the currently connected device	Please retry. Contact Manhattan Associates if the issue persists. While the issue is being resolved, you may use the native OS capabilities to disconnect from the currently connected device.
"No available devices found. You must have at least 1 reader paired to your device." while trying to manage the available RFID readers	Ensure that the device that you want to connect to has been successfully paired to your Store device and then try to connect to it using the Store app.

Custom Screens

Retailers can use the RFID reader on custom screens and base screens if the base screen does not, by default, support it. Refer to the Store RFID - Custom screen guide for additional information.

Custom Devices

Retailers can use other than base-certified RFID readers to read the RFID tags in the store application. Refer to the Store RFID - Custom Devices guide for additional information.

Unit Inventory Events

Below is the list of events and the corresponding EPC status update that occurs.

Event Name	EPC Status updated to
ShippedToAddress	Departed
Picked	Reserved
Packed	Reserved
CancelPack	Available
ShippedToStore	InBound
ShipConfirmation	InBound
Received	Available
MissingDuringReceive	Missing
StoreSale	Departed
Adjustment	Available
Removed	Removed
StoreReturn	Available
StorePickup	Departed
Overhead	Pending Receipt/Unexpected
EncodeTag	Available

UIs and Entities

The configuration UIs mentioned in this document are listed in the following table. This information can be used to navigate to API documentation and also specifies whether the configuration can be overridden at the location level.

UI Name	Component	Group	Seed Data Present?	Seed Data Profile	Profile Purpose	Is Profile Purpose Location Overrideable?
Store RFID Parameters	Store Facade	Store Config Services	Yes	Base	sto::storeConfig	Yes
Store RFID Configuration	Store RFID	RF Parameters	Yes	Base	srf::rfConfig	Yes
Store Fulfillment Parameters	Fulfillment	Fulfillment Parameters	Yes	Base	ful::fulfillmentConfig	Yes
RFID Reader Settings MUP UI	Device	Manage Devices	Yes	Base	dvc::rfidReaders	Yes